

CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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9231/21

May/June 2022

2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

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[Turn over

- 1 The curve C has polar equation $r = e^{\frac{3}{4}\theta}$ for $0 \leq \theta \leq \alpha$.

Given that the length of C is s , find α in terms of s .

[5]

[illegible]

- 3** The variables t and x are related by the differential equation

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} + x = t^2 + 1.$$

- (a) Find the general solution for x in terms of t . [6]

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- (b) Deduce an approximate value of $\frac{d^2x}{dt^2}$ for large positive values of t . [2]

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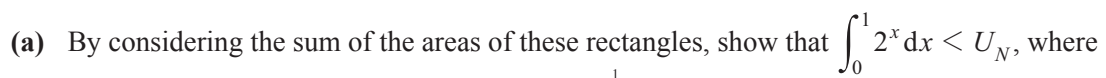
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[illegible]

- (b)** Use a similar method to find, in terms of N , a lower bound L_N for $\int_0^1 2^x dx$. [4]

This image shows a full page of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

- (c) Find the least value of N such that $U_N - L_N < 10^{-4}$. [2]

This image shows a full page of white paper with ten evenly spaced horizontal dashed lines, typical of primary school writing paper. The lines are thin and black, extending across the entire width of the page. There are no margins, text, or other markings present.

- 5 The variables x and y are such that $y = 0$ when $x = 0$ and

$$(x+1)y + (x+y+1)^3 = 1.$$

- (a) Show that $\frac{dy}{dx} = -\frac{3}{4}$ when $x = 0$. [3]

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- (b) Find the Maclaurin's series for y up to and including the term in x^2 . [7]

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- 6** Use the substitution $y = vx$ to find the solution of the differential equation

$$x \frac{dy}{dx} = y + \sqrt{9x^2 + y^2}$$

for which $y = 0$ when $x = 1$. Give your answer in the form $y = f(x)$, where $f(x)$ is a polynomial in x .
[10]

$$\operatorname{cosec} 7\theta = \frac{\operatorname{cosec}^7 \theta}{7 \operatorname{cosec}^6 \theta - 56 \operatorname{cosec}^4 \theta + 112 \operatorname{cosec}^2 \theta - 64}. \quad [6]$$

This image shows a full page of a document template designed for handwriting practice or general writing. It consists of approximately 20 evenly spaced horizontal dotted lines across the entire width of the page. The background is plain white, and there are no margins, headers, footers, or other markings present.

(b) Hence obtain the roots of the equation

$$x^7 - 14x^6 + 112x^4 - 224x^2 + 128 = 0$$

in the form $\operatorname{cosec} q\pi$, where q is rational. [5]

[illegible]

$$5x - y = 0,$$

$$x + 3y + 2z = 0,$$

does not have a unique solution. [2]

[illegible]
$$\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 \\ 5 & -1 & 0 \\ 1 & 3 & 2 \end{pmatrix}.$$

(b) Find a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A}^2 = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. [7]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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(c) Use the characteristic equation of $\mathbf{A}$ to show that

$$(\mathbf{A} + 6\mathbf{I})^2 = \mathbf{A}^4(\mathbf{A} + b\mathbf{I})^2,$$

where b is an integer to be determined.

[4]

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[illegible]

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